Project Proposal for OneDelivery: Agentic AI for Customer Service

GC Architecting AI
Systems Practice Module

Team 16

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1. Overview

1.1 Introduction

In the on-demand food delivery sector, speed is the primary currency. Customer satisfaction relies on immediate resolution when things go wrong. However, the industry currently faces a "Service Dilemma": the available support solutions are either too capable but expensive (humans) or too cheap but ineffective (basic chatbots).

This project proposes the design and implementation of an **Agentic AI System**. Unlike standard bots that simply recite information, this system is an "Action-Engine" capable of autonomously resolving complex customer service issues—from real-time driver tracking to financial dispute resolution—without human intervention.

1.2 Problem Statement

Current customer service models are failing to meet the demands of the modern "instant economy" due to two critical limitations:

1.2.1 The Rigidity of Traditional Chatbots

Standard chatbots operate on pre-defined scripts (decision trees). They lack **autonomy**. They can answer *"What is the refund policy?"* but they cannot execute the action *"Check the policy and refund me."* Because they cannot interact with backend systems (like payment gateways), they are "informational, not functional." If a customer's issue falls outside the pre-programmed script, the bot hits a dead end, forcing a transfer to a human and frustrating the user.

1.2.1 The Scalability Trap of Human Support

While human agents possess the autonomy to solve complex problems, they are **operationally unscalable**. Human support costs scale linearly: doubling the ticket volume requires doubling the staff. During peak hours (e.g., Friday dinner rush), it is impossible to hire enough staff to maintain low wait times. Relying on humans for routine tasks creates bottlenecks, high operational costs, and long resolution times that drive customer churn.

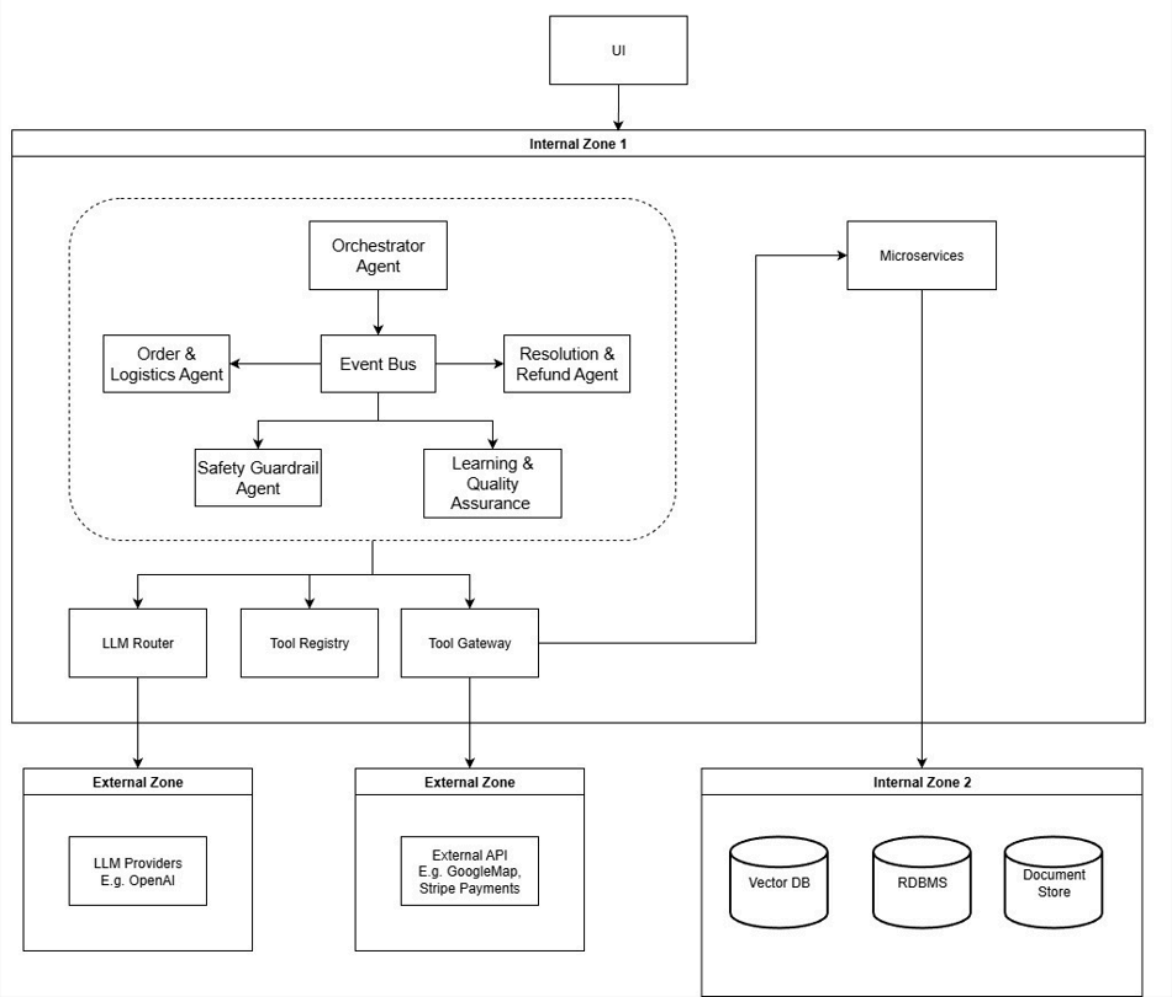
1.3 Proposed Solution

To bridge this gap, we present a **Multi-Agent System (MAS)** that combines the low operational cost of software with the execution capabilities of human agents.

The system utilizes a **Supervisor-Worker Architecture**. This design moves beyond simple conversation; it acts as an intelligent workforce.

- **It Perceives:** It understands complex, messy user intents.
- **It Reasons:** It decides which policy applies to the specific situation.
- **It Acts:** It uses software "tools" to perform real-world actions.

2. General Architecture



2.1 System Architecture Overview

The **OneDelivery** platform utilizes a **Multi-Agent Orchestration (MAO)** framework, moving away from rigid, linear chatbots toward a fleet of specialized, autonomous AI agents. This architecture is designed to handle the dynamic complexities of modern logistics—balancing speed, customer satisfaction, and operational safety.

2.1.1 Core Components

The architecture is structured into three functional layers:

- **The Brain (Orchestrator):** Acts as the central nervous system, to identify customer intent, manage multi-turn dialogue, and route tasks to the appropriate specialist.
- **The Specialists (Domain Experts):** Two dedicated agents—**Order & Logistics** and **Resolution & Refund**—interact with live delivery APIs and internal financial policies to execute real-time tracking and automated compensations.
- **The Oversight (Guardrails & QA):** A dual-layer protection system. The **Safety Agent** provides real-time monitoring to prevent policy violations, while the **QA Agent** performs post-interaction analysis to transform conversation logs into actionable business intelligence.

3. Scope of Work

Agent	Role	Responsibilities
Orchestrator Agent	Workflow Coordination	Identifies customer intent (delay, missing items, refund, complaint) Collects order details and user context Answers FAQs instantly Detects abusive language, threats, or harassment Routes issues to the appropriate internal agent Receive the response from the internal agents and translate it into a natural language response for the user
Order & Logistics Agent	Delivery lifecycle intelligence	Real-time order tracking Predicts delays using historical and live data Detects delays, rider issues, or route disruptions Coordinates rerouting, cancellation, or rescheduling
Resolution & Refund Agent	Automated problem resolution	Image Analysis (Verification) Issues refunds, credits, or replacements Applies compensation policies automatically Determines refund amounts based on SLA breaches
Guardian Agent	Ensures safe, compliant, and trustworthy interactions	Monitors all agent actions and customer interactions Prevents policy violations (over-refunding, data leakage) Enforces PII protection and compliance Blocks hallucinations or unsafe actions by other agents Triggers human escalation for high-risk scenarios
Quality Assurance Agent	Continuous improvement and optimization	Analyzes conversation outcomes, and escalation triggers. Identifies and categorizes incident types (e.g., Logistics vs. Technical) and generates human-readable summaries. Transcribes interaction metadata into the database for long-term trend analysis and audit trails. Generate monthly incidents reports from incidents analysis.

4. Effort Estimates

Phase	Task	Effort
1 - Design	Architecture Design	2
	API Design	1
	Safety & Governance Design	1
	Data Model	1
2 - Setup	Environment Setup	1
	Mock Data Generation	1
3 - Agents	Order & Logistics Agent - Prompt Engineering - Location Access - Predictive Analytics - Decision Explanation - Risk & Escalation Handling	5
	Resolution & Refund Agent - Prompt Engineering - Issue Legitimacy Validation - Decision Explanation - Compensation Calculation - Execute Resolution - Risk & Escalation Handling	5
	Orchestrator Agent - Prompt Engineering - State Manager - Sentiment Detection - Semantic Routing - Supervisor-Worker Orchestration - Output Synthesis	5
	Guardian Agent - Prompt Engineering - Safety & Guardrails Layer - PII Redaction Middleware - Prompt Injection Detector - Hallucination Blocking - HITL Approval Workflows	5
	Quality Assurance Agent - Prompt Engineering - Contextual Summarization - Incident Data Analysis - Automated Reporting Engine	5

4 - Integrations	Order Microservice	2
	Logistics Microservice	2
	Payment Microservice	2
	Audits Microservice	2
	External API Integration	2
	Shared State Management	2
5 - Frontend	End User UI	4
	Admin User	4
6 - Testing	Unit Testing	5
	Agent Behavior Testing	1
	Fraud & Abuse Testing	1
	Load Testing	1
	Safety Scenario Testing	1
	Fairness Validation	1
	Security Review	1
7 - DevOps	Automated Testing Setup	1
	Deployment Setup	2
	Infrastructure Setup	1
	Version Control	1
	Monitoring and Alerting	1
	Logging and Auditing	1
8 - Submission	Final Report and Demonstration	5